Reef photogrammetry: field form

One form per transect. Pages 1–3 are filled in the field (print on waterproof paper or copy onto the slate); page 4 is the post-site report, filled in the same day. Method: Underwater Photogrammetry and Field transects pages of the SEASTRONG Field Processing Tools documentation.

1. Transect header

Study area (BZ / SC / FJ) Belize, Seychelles or Fiji		**Corridor (C1–C6)** Corridor number in the study area	
Transect ID e.g. BZ140327C1reef (automatic in Acrobat)	will autocompile	**Date** Survey date, as mm/dd/yyyy	
Operators Names of the people in the water		**Snorkel / scuba** How the photo passes were swum	
Sensor 1 ID (seaward) Serial number of the seaward pressure sensor		**Sensor 1 position on tape (m)** Distance along the tape, e.g. -1	
Sensor 2 ID (landward) Serial number of the landward pressure sensor		**Sensor 2 position on tape (m)** Distance along the tape, e.g. 31	
GPS point above 0 m end (lat, lon) One approximate point at the surface above 0 m		**Bearing 0 → 30 m (°)** Compass bearing from 0 m towards 30 m	
Camera spacing (cm) Measured distance between the two lens centres		**Swimming height (m)** Height of the rig above the bottom, ~1.5–2 m	
Time zone of watch / phone Clock used for depth times, e.g. UTC+12		**Tide (rising / falling / slack)** Tide stage during the survey	

2. Before going in the water

- [ ] Two cameras set to the same time (GoPro app); labelled A and B
- [ ] Cameras charged, memory cards empty, identical settings (Timelapse, 0.5 s, 5K, Linear, Protune, Flat, ISO 200, shortest shutter, fixed white balance, JPEG)
- [ ] Rig assembled; camera spacing measured and written above
- [ ] At least 6 scale bars (bar list with you), 30 m tape, depth gauge, compass, GPS/phone, slate, weights or pegs, colour tab (optional)
- [ ] Conditions OK: visibility > ~4 m, little surge, tide not changing fast
- [ ] **No strong caustics on the bottom** (otherwise wait for clouds or a lower sun)

3. Layout (in the water)

- [ ] Tape laid cross-shore, straight and secured: 0 m at sensor 1 (seaward), 30 m at sensor 2
- [ ] At least 6 scale bars on both sides, within ~2 m of the tape, varied orientation and elevation; no damage to live coral
- [ ] Tape positions where the habitat changes noted below

Habitat changes along tape (m: description) e.g. 0–8 m coral; 8–14 m sand; 14–30 m coral	

4. Scale-bar targets: three depth readings at the centre of each target

Target code: printed on the target. **Position**: distance along the tape (m). **Side**: L or R looking landward. **Depth 1–3**: three readings a few seconds apart, gauge held at the target centre. **Time**: of the readings, for the tide correction.

Target code	Position on tape (m)	Side (L/R)	Depth 1 (m)	Depth 2 (m)	Depth 3 (m)	Time
e.g. T01	*2.5*	*L*	*1.62*	*1.60*	*1.63*	*10:42*

5. Depth profile: depth and time every 2 m, at each sensor, and 5–10 m beyond the 30 m mark

Depth from the gauge or a graduated stick held on the bottom next to the tape; time of each reading, for the tide correction. Write the position of the sensors in the last two rows.

Tape (m)	Depth (m)	Time
0		
2		
4		
6		
8		
10		
12		
14		
16		
18		
20		

Tape (m)	Depth (m)	Time
22		
24		
26		
28		
30		
32		
34		
36		
38		
40		
sensor 1		
sensor 2		

6. Photo survey

- [ ] Both cameras started as close to simultaneously as possible; colour tab in the first photos (if used)
- [ ] Pass 1: forward, camera A over the tape, ~1.5–2 m above the bottom, ~0.3 m/s
- [ ] Pass 2: backward, camera A over the tape
- [ ] Pass 3: zigzag, ~3 m each side, ~5 crossings each way
- [ ] Photos checked on both cameras: complete, sharp, not fogged; passes repeated if needed
- [ ] All scale bars (checked against the bar list) and the tape recovered; slate photographed

Cameras start time Time both cameras were started		**Cameras end time** Time both cameras were stopped	
Passes repeated? (which, why) e.g. pass 1 repeated, housing fogged			

Post-site report

Filled in the same day, after downloading the data. Send it with the data to the data manager.

A. Survey

Transect ID e.g. BZ140327C1reef (automatic in Acrobat)	will autocompile	**Date** Survey date, as mm/dd/yyyy	
Filled in by Who fills in this report		**Survey complete (Y / N)** All passes, targets and depths done	
Time in the water (h) From entering to leaving the water		**Visibility (m)** Estimated horizontal visibility	
Waves / surge Calm, moderate or strong		**Cloud / caustics** Overcast or sunny; caustics yes or no	
Current None, weak or strong			

B. Data

Photos camera A (n) Number of photos downloaded from camera A		**Photos camera B (n)** Number of photos downloaded from camera B	
Folder name camera A Transect ID + _camA		**Folder name camera B** Transect ID + _camB	
Target depths complete (Y / N) Three readings and time for every target		**Depth profile complete (Y / N)** Every 2 m, both sensors and beyond 30 m	
Slate photo saved (Y / N) Photo of the slate saved with the data		**Field sheet copied (Y / N)** Slate data typed into this form	

Data download link Link to the shared folder with photos, slate photo and this form	

C. Quality check (tick if a problem)

- Blurred or dark photos
- Fogged housing
- Caustics in part of the survey
- A camera stopped during a pass
- Targets moved or not visible
- Tape not straight or moved
- Missing depth readings

D. Deviations from the protocol, problems and notes